$N_{2b\,\text{finite}}$ (Group I, Row III): plus prescription and the p^+ integration, step by step

1 Starting expression

The expression below has the cutoffs Λ on the longitudinal integrals:

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{ \\
&e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \\
&\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}} \\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right] \\
&-\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right] \\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right] \\
&-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg] \\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y) \\
&-\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2} \\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]} \\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\frac{k^+}{q^+}\times \\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y) \\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]} \\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right] \\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.1}
$$

2 Step 0: shorthand

$$
\mathcal{N}\equiv\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+},\qquad \frac{\mathcal{N}}{2\pi}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+},
\tag{2.1}
$$

$$
\Phi(p^+)=e^{-ik\cdot(z-x)\frac{p^+}{k^+}},\qquad D(p^+)=p^+(y-x)^2+(k^+-p^+)(y-z)^2,\qquad T^{im}=\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2},
\tag{2.2}
$$

$$\Psi(q^+) = e^{-ik\cdot(z-x)\frac{k^+-q^+}{k^+}}, \qquad \widetilde{D}(q^+) = (k^+-q^+)(y-x)^2 + q^+(y-z)^2, \tag{2.3}$$

$$B^{im} = \frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}, \qquad A^{im} = \frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}, \tag{2.4}$$

$$\mathcal{C}_x = \big[U^{ab'}(x') - U^{ab'}(w')\big]\big[N_c U^{af}(y) - N_c U^{af}(x)\big]\rho^{b'}(x')\rho^f(y), \qquad \mathcal{C}_z = \mathcal{C}_x\big|_{U(x)\to U(z)}, \tag{2.5}$$

$$K_S = \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}. \tag{2.6}$$

3 Step 1: which terms are singular at the endpoints

In the p^+ block, Φ/D is finite at $p^+ = 0$ (where $D = k^+(y-z)^2$) and at $p^+ = k^+$ (where $D = k^+(y-x)^2$). The coefficients $\frac{p^+(k^+-p^+)}{k^{+2}}$, $\frac{k^+-p^+}{k^+}$, $\frac{p^+}{k^+}$ and 1 are finite. Only three structures are singular:

- $-\frac{k^+-p^+}{p^+}B^{im}$ in the p^+ block, at $p^+ \to 0$;
- $\frac{k^+}{q^+}A^{im}$ in the q^+ block, at $q^+ \to 0$ (that is, $p^+ \to k^+$);
- $\frac{p^+}{k^+-p^+}$ and $\frac{k^+-p^+}{p^+}$ in the subtraction term.

In every other term $\Lambda \to 0$ is taken directly, with an error of $O(\Lambda/k^+)$.

4 Step 2: split the singular coefficient

$$-\frac{k^+-p^+}{p^+}\,B^{im} = -\frac{k^+}{p^+}\,B^{im} + B^{im}. \tag{4.1}$$

The $+B^{im}$ piece is finite, so $\Lambda \to 0$ can be taken in it.

5 Step 3: plus prescription at $p^+ \to 0$

Definition:

$$\int_0^{k^+} dp^+ \Big[\frac{k^+}{p^+}\Big]_+ f(p^+) \equiv \int_0^{k^+} dp^+\,\frac{k^+}{p^+}\big[f(p^+) - f(0)\big]. \tag{5.1}$$

Add and subtract $f(0)$ (this is exact):

$$\begin{aligned}\int_\Lambda^{k^+-\Lambda} dp^+\frac{k^+}{p^+}f(p^+) &= \int_\Lambda^{k^+-\Lambda} dp^+\frac{k^+}{p^+}\big[f(p^+)-f(0)\big] + k^+f(0)\log\frac{k^+-\Lambda}{\Lambda} \\ &\xrightarrow{\Lambda\to 0} \int_0^{k^+} dp^+\Big[\frac{k^+}{p^+}\Big]_+ f(p^+) + k^+f(0)\log\frac{k^+}{\Lambda},\end{aligned} \tag{5.2}$$

with $f(p^+) = \Phi(p^+)\,T^{im}B^{im}/D(p^+)$.

6 Step 4: evaluate $k^+f(0)$

At $p^+ = 0$: $\Phi = 1$ and $D = k^+(y-z)^2$, so $k^+f(0) = T^{im}B^{im}/(y-z)^2$. Contract the two terms of B with T:

$$\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2}\,\frac{(y-z)^i(x-z)^m}{(x-z)^2} = -\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(x-z)^2}, \tag{6.1}$$

$$\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2}\,\frac{(y-x)^m(y-z)^i}{2(y-x)^2} = -\frac{(x'-w')\cdot(y-z)\,(x-z)\cdot(y-x)}{2(x'-w')^2(x-z)^2(y-x)^2}. \tag{6.2}$$

Hence

$$k^+f(0) = -\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\left[\frac{1}{(x-z)^2} + \frac{(x-z)\cdot(y-x)}{2(x-z)^2(y-x)^2}\right]. \tag{6.3}$$

Multiplying by the -1 of Step 2 and by $\frac{1}{2\pi}$ from $\frac{dp^+}{2\pi}$ gives the endpoint term of the p^+ block:

$$E_p = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\left[\frac{1}{(x-z)^2} + \frac{(x-z)\cdot(y-x)}{2(x-z)^2(y-x)^2}\right]\mathcal{C}_x. \tag{6.4}$$

7 Step 5: the q^+ block

The two q^+ integrals of (1.1) together carry $\big(\frac{k^+}{q^+}-1\big)A^{im}$.

- The -1 piece is finite, so $\Lambda\to 0$ gives $-\int_{k^+}^0 dq^+(\cdots)A^{im}$.
- For the $\frac{k^+}{q^+}$ piece, use $\int_{k^+-\Lambda}^{\Lambda}=-\int_{\Lambda}^{k^+-\Lambda}$ and Step 3 with $h(q^+)=\Psi(q^+)\,T^{im}A^{im}/\widetilde{D}(q^+)$:

$$\int_{k^+-\Lambda}^{\Lambda} dq^+\frac{k^+}{q^+}h(q^+)=\int_{k^+}^{0} dq^+\Big[\frac{k^+}{q^+}\Big]_+h(q^+)-k^+h(0)\log\frac{k^+}{\Lambda}. \tag{7.1}$$

At $q^+=0$: $\widetilde{D}=k^+(y-x)^2$ and $e^{-ik\cdot(w'-z)}\Psi(0)=e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)}=e^{-ik\cdot(w'-x)}$. The contraction is

$$T^{im}A^{im}=\frac{(x'-w')\cdot(y-x)}{(x'-w')^2}\left[-\frac{1}{(x-z)^2}+\frac{(x-z)\cdot(y-z)}{2(x-z)^2(y-z)^2}\right], \tag{7.2}$$

so the endpoint term of the q^+ block is

$$E_q=\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-x)}\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\left[\frac{1}{(x-z)^2}-\frac{(x-z)\cdot(y-z)}{2(x-z)^2(y-z)^2}\right]\mathcal{C}_x. \tag{7.3}$$

8 Step 6: the subtraction term, integrated exactly

The kernel K_S is independent of p^+, so the p^+ integral is done exactly. With $\xi=p^+/k^+$ and $\lambda=\Lambda/k^+$:

$$\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}dp^+\Big[\tfrac{p^+(k^+-p^+)}{k^{+2}}+\tfrac{p^+}{k^+-p^+}+\tfrac{k^+-p^+}{p^+}\Big]=\int_{\lambda}^{1-\lambda}d\xi\Big[\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big], \tag{8.1}$$

$$\int_{\lambda}^{1-\lambda}d\xi\,\xi(1-\xi)=\frac{1}{6}+O(\lambda^2), \tag{8.2}$$

$$\int_{\lambda}^{1-\lambda}d\xi\,\frac{\xi}{1-\xi}=\int_{\lambda}^{1-\lambda}d\xi\Big[\frac{1}{1-\xi}-1\Big]=\log\frac{1-\lambda}{\lambda}-(1-2\lambda), \tag{8.3}$$

$$\int_{\lambda}^{1-\lambda}d\xi\,\frac{1-\xi}{\xi}=\int_{\lambda}^{1-\lambda}d\xi\Big[\frac{1}{\xi}-1\Big]=\log\frac{1-\lambda}{\lambda}-(1-2\lambda), \tag{8.4}$$

$$\text{sum}=2\log\frac{k^+-\Lambda}{\Lambda}-\frac{11}{6}+O(\lambda)\ \to\ 2\log\frac{k^+}{\Lambda}-\frac{11}{6}. \tag{8.5}$$

The plus-prescription form gives the same result. Write $\frac{\xi}{1-\xi}=\big[\frac{1}{1-\xi}\big]_+-1$ and $\frac{1-\xi}{\xi}=\big[\frac{1}{\xi}\big]_+-1$. The kernel is ξ-independent and $\int_0^1[\cdot]_+=0$, so only $\frac{1}{6}-1-1=-\frac{11}{6}$ survives; the log's are the endpoint pieces.

Including $\frac{1}{2\pi}$, the subtraction term becomes

$$\begin{aligned}&-\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\Big[2\log\frac{k^+}{\Lambda}-\frac{11}{6}\Big]\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}K_S\,\mathcal{C}_z\\&=\underbrace{+\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\frac{11}{6}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}K_S\,\mathcal{C}_z}_{\text{the }11/6\text{ term}}\ \underbrace{-\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}2\log\frac{k^+}{\Lambda}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}K_S\,\mathcal{C}_z}_{E_S}.\end{aligned} \tag{8.6}$$

9 Step 7: result

$$\begin{aligned}\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k}=&\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\&\left.-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\right.\end{aligned}$$

$$
\begin{aligned}
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\;\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left\{\left[\frac{k^+}{q^+}\right]_+-1\right\}\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{11}{6}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&+E_p+E_q+E_S,
\end{aligned}
\tag{9.1}
$$

where

$$
\begin{aligned}
E_p+E_q+E_S=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\frac{k^+}{\Lambda}\int_{x,y,z,x',w'}\Bigg\{&e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\left[\frac{1}{(x-z)^2}+\frac{(x-z)\cdot(y-x)}{2(x-z)^2(y-x)^2}\right]\mathcal{C}_x\\
&+e^{-ik\cdot(w'-x)}\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\left[\frac{1}{(x-z)^2}-\frac{(x-z)\cdot(y-z)}{2(x-z)^2(y-z)^2}\right]\mathcal{C}_x-2\,e^{-ik\cdot(w'-z)}K_S\,\mathcal{C}_z\Bigg\}.
\end{aligned}
\tag{9.2}
$$

10 Where each piece goes

- The Λ-independent part of (9.1) (everything except the blue terms) is exactly the $N_{2b\,\text{finite}}$ equation of the file `Other_terms_updated.tex`, including the $\frac{11}{6}$ term.
- The blue $\log(k^+/\Lambda)$ terms $E_p+E_q+E_S$ are the "$N_{2b\,\text{finite}}$" entries of the JIMWLK file. They do not belong in the finite file.
- $E_S=-2\log\frac{k^+}{\Lambda}\,K_S\,\mathcal{C}_z$ cancels the $+2\log\frac{k^+}{\Lambda}\,K_S\,\mathcal{C}_z$ of $N_{2b\,\text{UV}}$. In the JIMWLK file these appear as the $+2N_c$ and $-2N_c$ blocks, which are equal and opposite.
- The $\frac{11}{6}$ term is kept unintegrated over x. It diverges at $x\to z$ and is the subtraction partner of the remaining terms, which diverge oppositely there; their sum is UV finite. Its partner in $N_{2b\,\text{UV}}$ is the $-\frac{11}{6}$ that cancels the $\frac{11}{3}$ of Row II.